Posts and Telecommunications Institute of Technology



Object-Oriented Programming (OOP) Introduction

Faculty: Information Technology 1

Department: Computer Science

Instructor: Pham Hoang Viet

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Overview

- Course code: INT1332_CLC
- Number of credits: 3
- Course type: Compulsory
- Instructors: Pham Hoang Viet (vietph@ptit.edu.vn, 0945296388)
  Nguyen Dinh Quang (ndquang@ptit.edu.vn, 0971824607)

Objectives

- (Knowledge) The aim of this course is to provide learners the important knowledge of object oriented-programming method, including:
  - Fundamental knowledge about object-oriented methodology.
  - Knowledge of Java programming language, object oriented programming in Java.
  - Knowledge about Exception handling techniques, programming Graphical User Interface, Generics and Collections in Java.
- (Skills) The aim of this course is to equip learners in solving a real problem with skills in:
  - Ability to write multi-class programs with the Java programming language following object-oriented principles.
  - Ability to coordinate and work in groups

References

- [1] **Bài giảng Lập trình hướng đối tượng**, Nguyễn Mạnh Sơn, 2020
- [2] Y. Daniel Liang, **Introduction to Java Programming**, 11th edition, 2019
- [3] **The Java Tutorials**, Oracle Java Documentation (https://docs.oracle.com/javase/tutorial/)
- [4] **https://code.ptit.edu.vn**

  Account:

  Username: Student code

  Default password : ddmmyyyy

Main contents

- ❖Java programming language
- ❖Object oriented programming in Java
- ❖Exception handling in Java
- ❖Files and IO
- ❖Programming Graphical User Interface (GUI) in Java
- ❖Generics and Collections in Java
- ❖Thread in Java

Course map

1. Introduction to Java programming language
2. Classes and Objects - Encapsulation
3. Arrays and Strings
4. Exception Handling
5. Files and IO
6. Inheritance and Polymorphism
7. Generics and Collections
8. Thread in Java
9. GUI and Event handler programming
10. Design patterns (extension – available if…)

Shared Folder: Download lecture slides + Upload project reports

https://drive.google.com/drive/folders/1wnqHvpPjwIjJl5Gm3MBb1VrQgjNCUwDl?usp=sharing



Grading

- **Attendance: 10%**
  - QR scan: At the start of class and during the break (if late, just wait; do not ask).
  - Full attendance = 8 points; bonus points for class participation.
  - Complete at least 50% of the exercises on https://code.ptit.edu.vn
- **Midterm Exam: 20%**
  - Written format
  - 60 minutes
- **Project: 20%**
  - Group: 3–4 members.Report file: Upload to Drive (deadline: end of Session 30).
  - Presentation: 10–15 minute presentation + Q&A.
- **Final Exam: 50%**
  - Lab

Requirement

❖**Attitude and Classroom Conduct:**

- Students are expected to attend class sessions regularly and complete all assigned exercises and written work.
- Please avoid excessive talking or behaviors that may distract the instructor or disrupt the learning environment.

❑Questions are welcome.

❖**Examinations and Assessments:**

- Cheating and copying others' work are not permitted.
- Please do not request grade adjustments.
- Assignments should be submitted by the stated deadlines.
- Failure to follow these guidelines may result in a score of zero for the relevant assessments.

Basic-level topics

#	Topic	Key subjects	Key OOP features
1	**Library Management System**	Book, Member, Librarian, Loan	Association, inheritance
2	**Student Management System**	Student, Course, Department, Grade	Composition, collections
3	**Bank Account Management**	Customer, Account, Transaction	Inheritance, polymorphism
4	**Hotel Room Management**	Room, Guest, Reservation	Association, enum
5	**Cinema Ticket Booking**	Movie, Theater, Show, Seat, Ticket	Composition
6	**Vehicle Rental System**	Vehicle, Car, Motorbike, Customer, Rental	Inheritance
7	**Gym Membership System**	Member, Trainer, Membership, Course	Association
8	**Restaurant Ordering System**	MenuItem, Table, Order, OrderItem	Composition
9	**Parking Lot Management**	Vehicle, ParkingSpot, Ticket	Polymorphism
10	**Personal Expense Manager**	User, Account, Transaction, Category	Collections, file I/O

Intermediate-level topics

#	Topic	Key subjects	Applicable patterns/OOP concepts
11	**University Course Registration**	đăng ký môn, prerequisite, lịch học	Strategy, Observer
12	**Hospital/Clinic Management**	bệnh nhân, bác sĩ, lịch khám, hóa đơn	Factory, Strategy
13	**Online Shopping System**	product, cart, order, payment	Strategy, Factory
14	**Food Delivery System**	restaurant, order, driver, delivery	State, Strategy
15	**Learning Management System**	course, lesson, assignment, grading	Factory, Observer
16	**Online Examination System**	question, exam, submission, grading	Factory, Strategy
17	**Event & Ticketing System**	event, ticket, booking, check-in	Factory, Observer
18	**Car Service Center**	vehicle, repair, technician, parts	Strategy, Factory
19	**Dormitory Management System**	student, room, contract, utility bill	State, Strategy
20	**Task & Project Management System**	project, task, member, comment	Observer, State

Advanced-level topics

#	Đề tài	Nội dung	OOP/Pattern nổi bật
21	**Airline Reservation System**	flight, aircraft, seat, passenger, booking	State, Strategy, Factory
22	**Logistics & Parcel Tracking System**	package, warehouse, shipment, tracking	State, Observer
23	**Ride-Hailing System**	passenger, driver, vehicle, trip, payment	Strategy, Observer
24	**Smart Home Management System**	sensors, devices, rules, automation	Observer, Command
25	**Smart Traffic Management System**	intersection, road, vehicle, traffic light	State, Observer
26	**Digital Wallet & Payment System**	wallet, transaction, payment, transfer	Strategy, Factory
27	**Insurance Management System**	policy, customer, claim, compensation	Strategy, State
28	**Warehouse & Inventory Management**	product, supplier, warehouse, order	Observer, Factory
29	**Game Tournament Management**	team, player, match, tournament, ranking	Strategy, State
30	**Food Delivery Marketplace**	restaurant, customer, driver, order, promotion	Strategy, State, Observer

The report and presentation slides should demonstrate...

Criteria	Points
Problem analysis & requirements	10
UML & object modeling	10
Encapsulation & class design	10
Association/composition	10
Inheritance & polymorphism	10
Interfaces & abstraction	10
Collections, generics, exceptions	10
SOLID/design patterns	10
Code quality & testing	10
Demo & presentation	10
Total	**100**

SOLID

Chữ	Nguyên lý	Ý nghĩa ngắn gọn
S	Single Responsibility Principle	Một class nên có **một trách nhiệm chính**
O	Open/Closed Principle	**Mở để mở rộng**, đóng để sửa đổi
L	Liskov Substitution Principle	Lớp con phải có thể **thay thế lớp cha**
I	Interface Segregation Principle	Không ép class phụ thuộc vào interface nó **không cần**
D	Dependency Inversion Principle	Phụ thuộc vào **abstraction**, không phụ thuộc implementation cụ thể

Mile stones

- **Milestone 1 – Problem Analysis** → mô tả requirements và use cases (Sau chương 1).
- **Milestone 2 – Object-Oriented Analysis** → xác định objects/classes và trách nhiệm (Sau chương 3).
- **Milestone 3 – UML Design** → class diagram, relationships, inheritance (Sau chương 6).
- **Milestone 4 – Core Implementation** → Java classes + collections + exception handling (Sau chương 8).
- **Milestone 5 – Design Improvement** → SOLID + refactoring + design patterns (Sinh viên tự làm, giảng viên không giám sát).
- **Milestone 6 – Final Product** → testing + demo + presentation (Buổi thuyết trình).

Choosing Your Project: What to Discuss With Your Team

Problem fit: does the idea naturally use the topics from this chapter — variables, control structures, and console I/O — without requiring knowledge you haven't learned yet?
Scope: can a small team realistically design, code, and test it within the semester timeline? Prefer a small idea done well over an ambitious idea left unfinished.
Data to manage: what will your program store and manipulate (student records, book catalog, quiz questions, expenses)?
User interaction: will the user interact through a simple text menu on the console for now, evolving toward a GUI later in the course?
Team roles: who owns which class or feature, and how will you integrate everyone's work before the deadline?



Deliverable: proposal naming your application, its core menu options, analysis, design — due one week after reminded.